

Rohithkumar Saravanan

Limerick, Ireland | 089 201 5368 | rohithrk.in@gmail.com | LinkedIn: www.linkedin.com/in/rohith-rk

PROFESSIONAL SUMMARY

AI/ML master's student at the University of Limerick with hands-on experience building deep learning and computer vision systems for medical imaging and biomedical signal analysis. Microsoft Certified in Fabric data engineering (DP-700) and SQL AI development (DP-800), with additional training in AWS machine learning and generative deep learning. Published researcher and Best Paper Award recipient with a track record of award-winning academic projects.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C/C++, JavaScript, SQL, HTML/CSS

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, Image Processing, Generative Deep Learning (TensorFlow)

Cloud & Data Platforms: Microsoft Fabric, Azure SQL / AI-enabled database solutions, AWS

Tools & Version Control: Git, GitHub

Web Development: Front-End Development (HTML, CSS, JavaScript)

Professional Skills: Communication, Project Management, Leadership, Team Work, Time Management, Adaptability

EDUCATION

University of Limerick 2025 – 2026

Master's in Artificial Intelligence and Machine Learning

SRM Institute of Science and Technology 2021 – 2025

Bachelor of Technology (B.Tech) in Electronics and Computer Engineering — CGPA: 8.34

PROJECTS

Advanced Surface EMG Analysis for ALS Neuromuscular Dysfunction Mar 2025

- Designed and built Arduino-based sEMG signal acquisition hardware.
- Developed a preprocessing pipeline and deep learning models (CNN, LSTM, CNN-LSTM) to classify EMG signals.
- Built a web interface for real-time prediction and visualization.

Enhancing Chest CT Scan Infection Type Classification using BEMD Apr 2024

- Proposed a new Bidimensional Empirical Mode Decomposition (BEMD)-based preprocessing method to improve CT scan image quality.
- Built a deep learning model for multi-class classification of chest infection types.

Pneumonia Detection using PDNN Jul 2024

- Developed a dropout-regularized deep neural network (PDNN) architecture for pneumonia detection in CT scans.

Biorthogonal Wavelet Image Enhancement for Pneumonia Diagnosis Aug 2024

- Built a biorthogonal wavelet-based image enhancement pipeline for CT scan images.
- Achieved 99.20% classification accuracy using a DenseNet architecture; findings published (see Publications).

Deep Transfer Learning for Lung Cancer Classification Jul 2024

- Applied GoogLeNet transfer learning on the LIDC dataset to classify lung cancer from CT scans.

Triplet-Attention Residual Network for Pneumonia Detection Jul 2024

- Developed LuNet, a triplet-attention residual network, achieving 95.34% accuracy on CT scan datasets.

EXPERIENCE

Python Intern Aug 2023 – Nov 2023

Shiash Info Solutions Private Limited

- Completed a hands-on internship in Python programming and applied problem-solving on the project “Estimation of Carbon for Industry and Household to Prevent Environment and Industry Progression.”
- Developed Python-based models to estimate carbon emissions from industrial and household sources.
- Applied data analysis techniques to support environmental sustainability and industrial efficiency.

Artificial Intelligence & Machine Learning Internship Apr 2024 – Jun 2024

Google for Developers (India Edu Program)

- Completed a structured 10-week program covering foundational AI and machine learning concepts.
- Worked on ML model training and optimization techniques to improve model accuracy and performance.
- Applied Google Developer ecosystem best practices to design scalable AI solutions.

Cyber Security Intern (Industrial Training) Mar 2023 – Apr 2023

Industrial Training Program

- Completed industry-mentored training in cybersecurity fundamentals, including threat analysis, system vulnerabilities, and security best practices.

CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate (DP-700)	<i>Jun 2026</i>
Microsoft Certified: SQL AI Developer Associate (DP-800)	<i>Jul 2026</i>
AWS Academy Graduate — AWS Academy Machine Learning Foundations	
Deloitte Data Analytics Job Simulation — Certificate of Completion	<i>Jul 2026</i>

PUBLICATIONS

- Biorthogonal Wavelet-Based Image Enhancement for Accurate Pneumonia Diagnosis from Lung CT Scans Using Deep Learning — Springer Link
- Advanced Surface EMG Analysis for ALS Neuromuscular Dysfunction Using Deep Learning Models (Submitted)

AWARDS

Best Paper Award — All India Two Days Seminar, The Institution of Engineers (India)	<i>Jul 2024</i>
Best Project Award — Multidisciplinary Domain, SRM Institute of Science and Technology	<i>Apr 2025</i>
3rd Place — COMSPRO 2025 Project Contest, SRM Institute of Science and Technology	<i>Apr 2025</i>

CONTINUING EDUCATION

- Generative Deep Learning with TensorFlow — Coursera
- Exam Prep: AWS Certified AI Practitioner (AIF-C01) — Coursera
- Supervised Machine Learning — Coursera
- Advanced Computer Vision with TensorFlow — Coursera
- Advanced Diploma in Python Programming — CSC Institute
- Introduction to Front-End Development — Coursera
- Programming with JavaScript — Coursera
- HTML and CSS in Depth — Coursera
- Version Control with Git — Coursera
- MATLAB Onramp — MathWorks
- Arduino Programming — KitFlix

WORKSHOPS AND SEMINARS

Signal Processing and Artificial Intelligence — The Institution of Engineers (India)	<i>Apr 2023</i>
Kuka Robotics Workshop — Chennai Institute of Technology	<i>Sep 2023</i>
Industrial Automation Workshop — Chennai Institute of Technology	<i>Oct 2023</i>
All India Two Days Seminar — The Institution of Engineers (India)	<i>Jul 2024</i>

POSITION OF RESPONSIBILITY

Committee Head — Directorate of Student Affairs (DSA)	<i>Jun 2021 – May 2025</i>
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